

DevOps Project 5 - Containerizing Spring Boot Application and Docker Image Scanning

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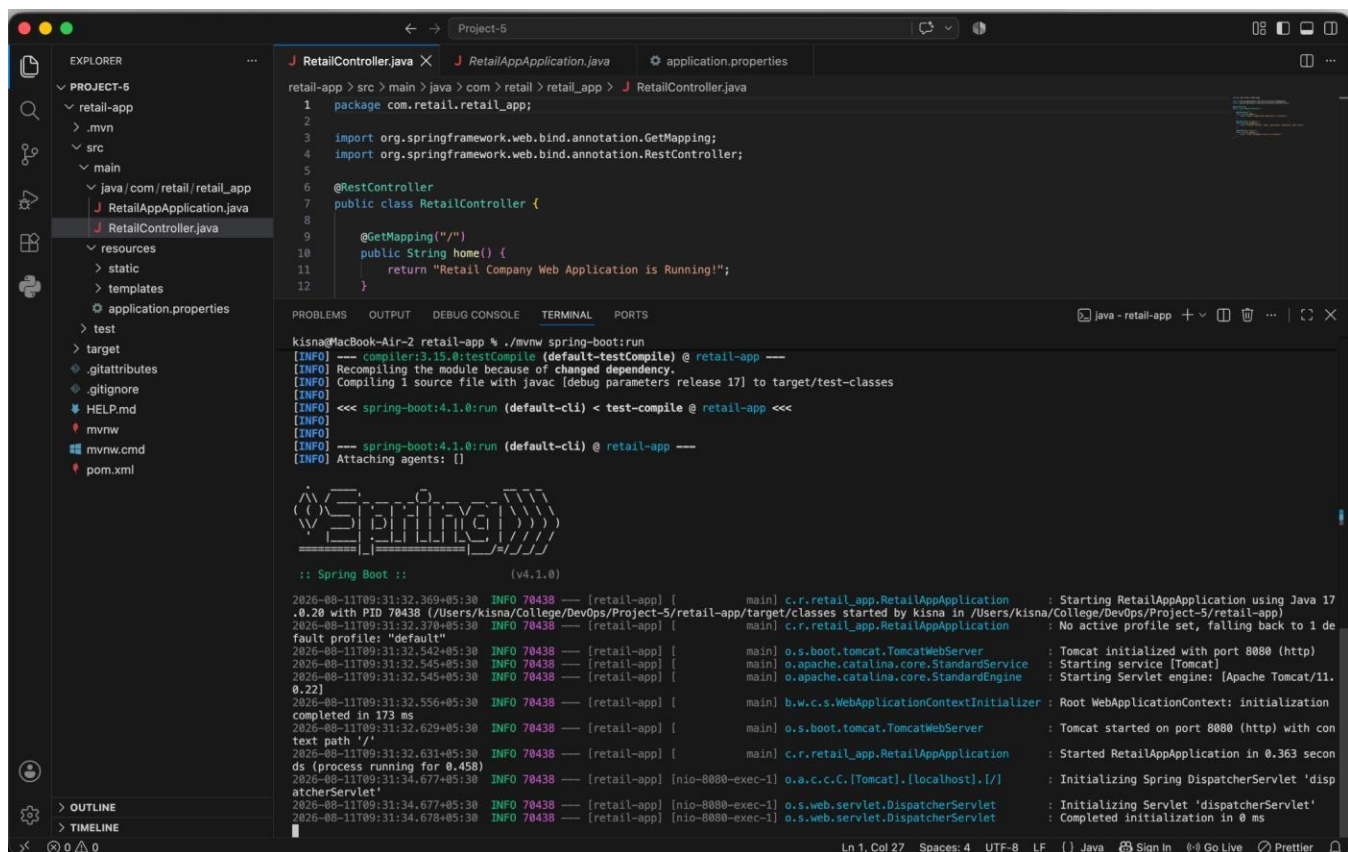
PRN.: 23070122150

Batch: 2023-27 Class:

A

1. Spring Boot Application Running Successfully

This screenshot shows the Spring Boot application starting successfully on Java 17 with the embedded Tomcat server running on port 8080.



```
Project-5
J RetailController.java X J RetailAppApplication.java application.properties
retail-app > src > main > java > com > retail > retail_app > J RetailController.java
1 package com.retail.retail_app;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class RetailController {
8
9     @GetMapping("/")
10    public String home() {
11        return "Retail Company Web Application is Running!";
12    }
13 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
kisna@MacBook-Air-2 retail-app % ./mvnw spring-boot:run
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ retail-app ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ retail-app <<<
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ retail-app ---
[INFO] Attaching agents: []

:: Spring Boot ::
(v4.1.0)

2026-08-11T09:31:32.369+05:30 INFO 70438 [retail-app] [main] c.r.retail_app.RetailAppApplication : Starting RetailAppApplication using Java 17
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] c.r.retail_app.RetailAppApplication : No active profile set, falling back to 1 de
fault profile: "default"
2026-08-11T09:31:32.542+05:30 INFO 70438 [retail-app] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-08-11T09:31:32.545+05:30 INFO 70438 [retail-app] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-08-11T09:31:32.545+05:30 INFO 70438 [retail-app] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.
0.22]
2026-08-11T09:31:32.556+05:30 INFO 70438 [retail-app] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization
completed in 173 ms
2026-08-11T09:31:32.629+05:30 INFO 70438 [retail-app] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with con
text path '/'
2026-08-11T09:31:32.631+05:30 INFO 70438 [retail-app] [main] c.r.retail_app.RetailAppApplication : Started RetailAppApplication in 0.363 secon
ds (process running for 0.458)
2026-08-11T09:31:34.677+05:30 INFO 70438 [retail-app] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'disp
atcherServlet'
2026-08-11T09:31:34.677+05:30 INFO 70438 [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-08-11T09:31:34.678+05:30 INFO 70438 [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
```

Figure 1: Spring Boot Application Running Successfully

2. Retail Web Application Tested in Browser

This screenshot verifies that the retail web application is accessible through the browser at localhost:8080 and returns the expected application message.

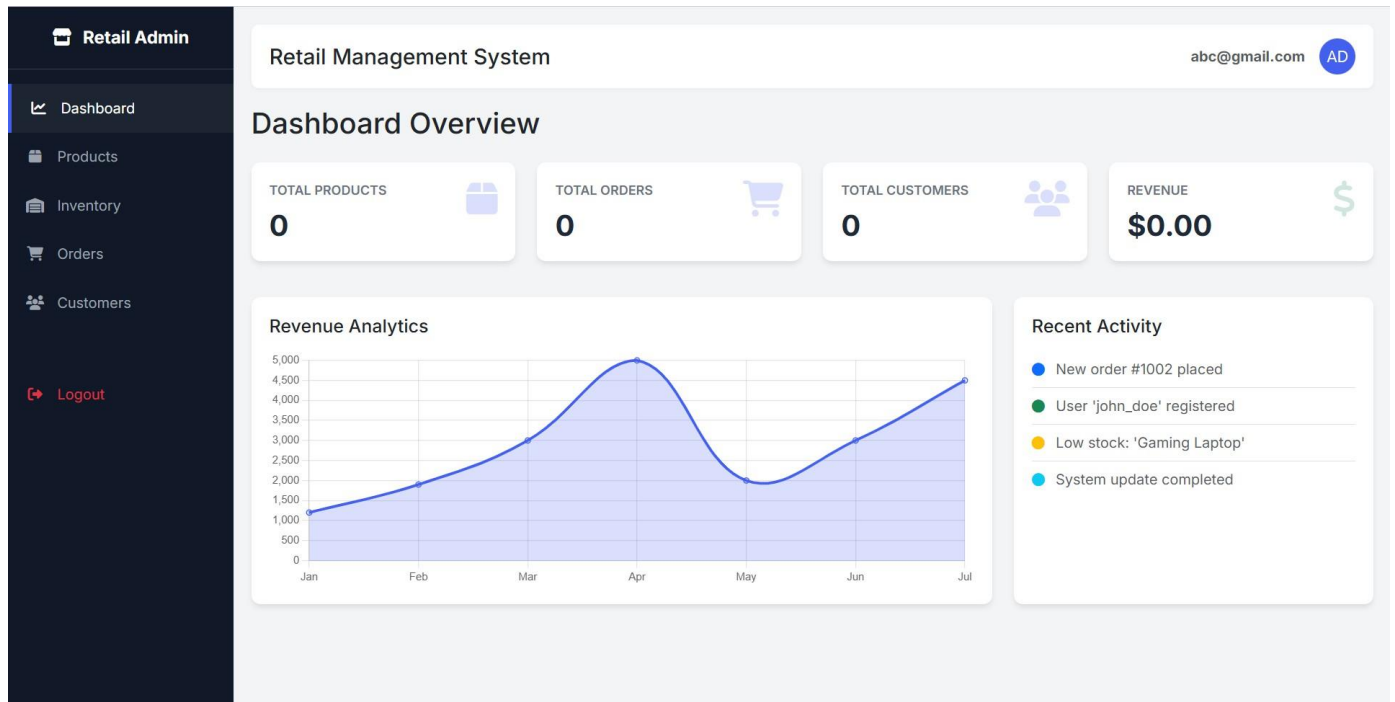
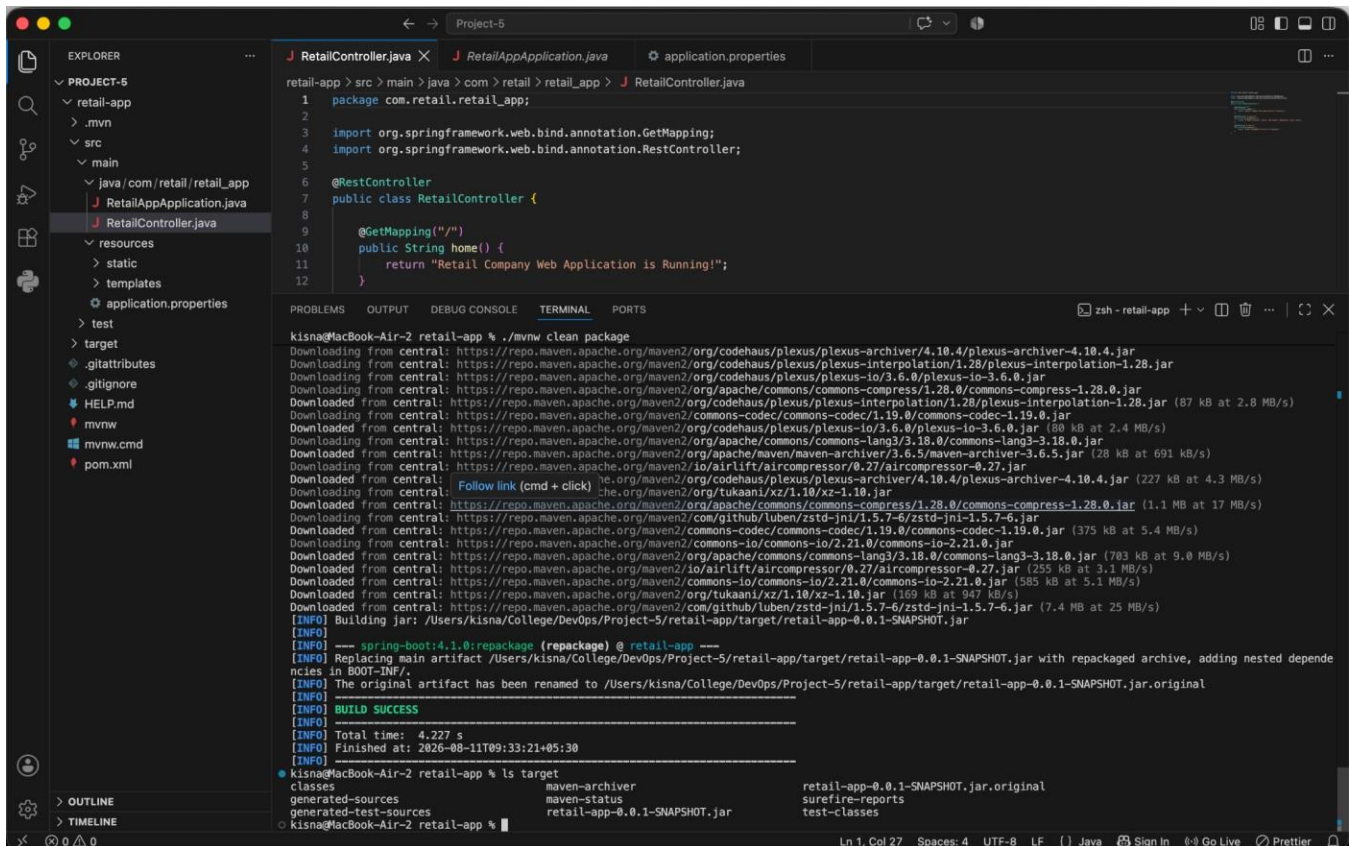


Figure 2: Retail Web Application Tested in Browser

3. Spring Boot JAR File Built Successfully

This screenshot shows Maven completing the package operation with BUILD SUCCESS and generating the retail-app-0.0.1-SNAPSHOT.jar file.



The screenshot displays an IDE interface with the following components:

- EXPLORER:** Shows the project structure for 'Project-5', including 'retail-app' with subdirectories 'main' and 'resources'. The file 'RetailController.java' is selected.
- Code Editor:** Displays the content of 'RetailController.java', which includes imports for Spring Framework annotations and a REST controller class.
- Terminal:** Shows the output of the command 'mvnw clean package'. It lists the download progress of various Maven dependencies from the central repository, followed by the successful execution of the build process.
- Output:** Displays the final build results, including the total time (4.227 s) and the completion time (2026-08-11T09:33:21+05:30).

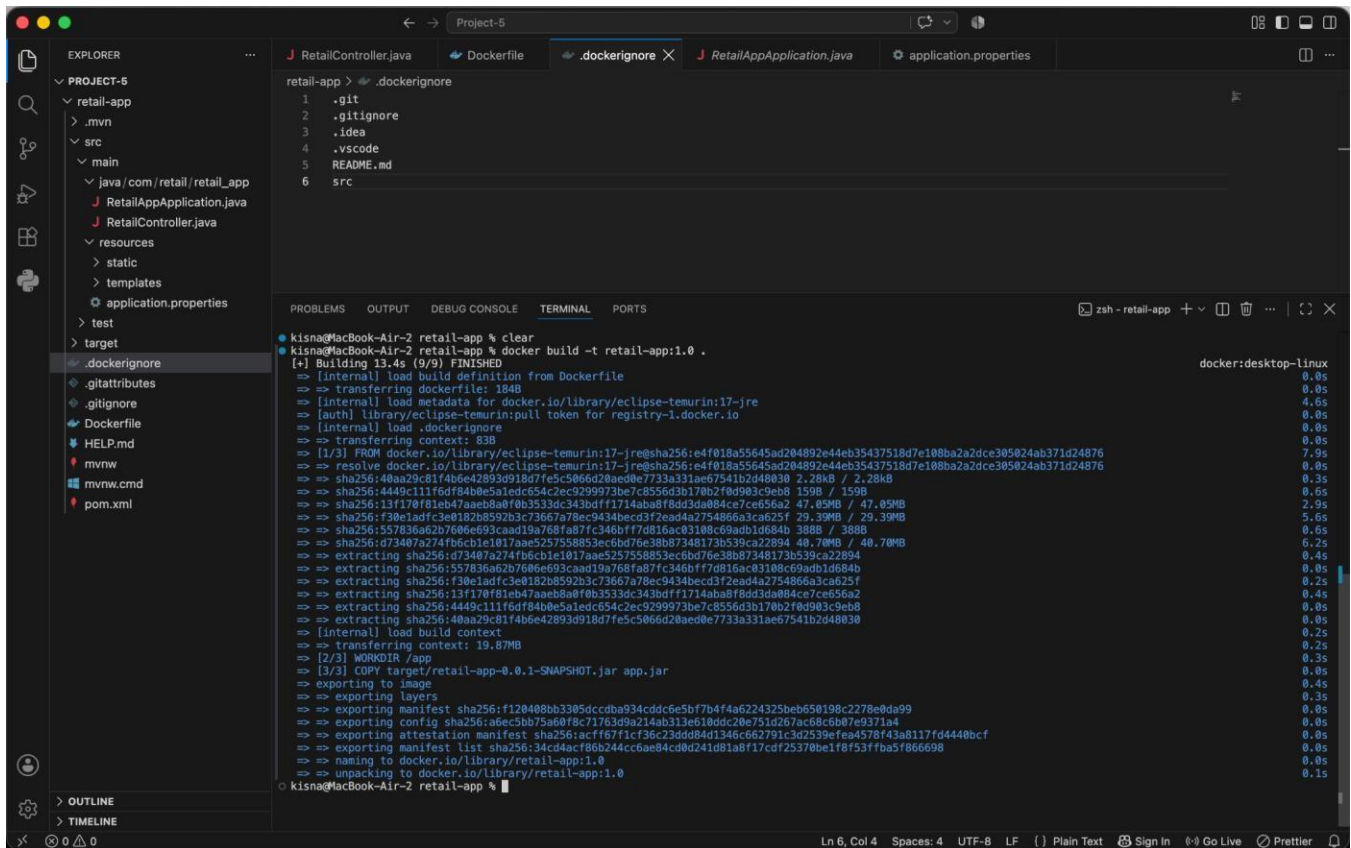
```
retail-app > src > main > java > com > retail > retail_app > J RetailController.java
1 package com.retail.retail_app;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class RetailController {
8
9     @GetMapping("/")
10    public String home() {
11        return "Retail Company Web Application is Running!";
12    }
13 }
```

```
kisna@MacBook-Air-2 retail-app % ./mvnw clean package
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.10.4/plexus-archiver-4.10.4.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.28/plexus-interpolation-1.28.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.6.0/plexus-io-3.6.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.28.0/commons-compress-1.28.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.28/plexus-interpolation-1.28.jar (87 kB at 2.8 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-codec/commons-codec/1.19.0/commons-codec-1.19.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.6.0/plexus-io-3.6.0.jar (88 kB at 2.4 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.18.0/commons-lang3-3.18.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.jar (28 kB at 691 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.jar (28 kB at 4.3 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.10.4/plexus-archiver-4.10.4.jar (227 kB at 4.3 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.28.0/commons-compress-1.28.0.jar (1.1 MB at 17 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.jar (28 kB at 5.4 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-codec/commons-codec/1.19.0/commons-codec-1.19.0.jar (375 kB at 5.4 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-io/2.21.0/commons-io-2.21.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.18.0/commons-lang3-3.18.0.jar (703 kB at 9.8 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-io/2.21.0/commons-io-2.21.0.jar (585 kB at 3.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-io/2.21.0/commons-io-2.21.0.jar (585 kB at 5.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.10/xz-1.10.jar (169 kB at 947 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.10/xz-1.10.jar (169 kB at 25 MB/s)
[INFO] Building jar: /Users/kisna/College/DevOps/Project-5/retail-app/target/retail-app-0.0.1-SNAPSHOT.jar
[INFO] --- spring-boot:4.1.0:repackage (repackage) @ retail-app ---
[INFO] Replacing main artifact /Users/kisna/College/DevOps/Project-5/retail-app/target/retail-app-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/
[INFO] The original artifact has been renamed to /Users/kisna/College/DevOps/Project-5/retail-app/target/retail-app-0.0.1-SNAPSHOT.jar.original
[INFO] BUILD SUCCESS
[INFO] Total time: 4.227 s
[INFO] Finished at: 2026-08-11T09:33:21+05:30
[INFO]
kisna@MacBook-Air-2 retail-app % ls target
classes                                maven-archiver                    retail-app-0.0.1-SNAPSHOT.jar.original
generated-sources                     maven-status                     surefire-reports
generated-test-sources                retail-app-0.0.1-SNAPSHOT.jar    test-classes
kisna@MacBook-Air-2 retail-app %
```

Figure 3: Spring Boot JAR File Built Successfully

4. Docker Image Successfully Built

This screenshot demonstrates the successful Docker build process using the Dockerfile and creation of the retail-app:1.0 image.



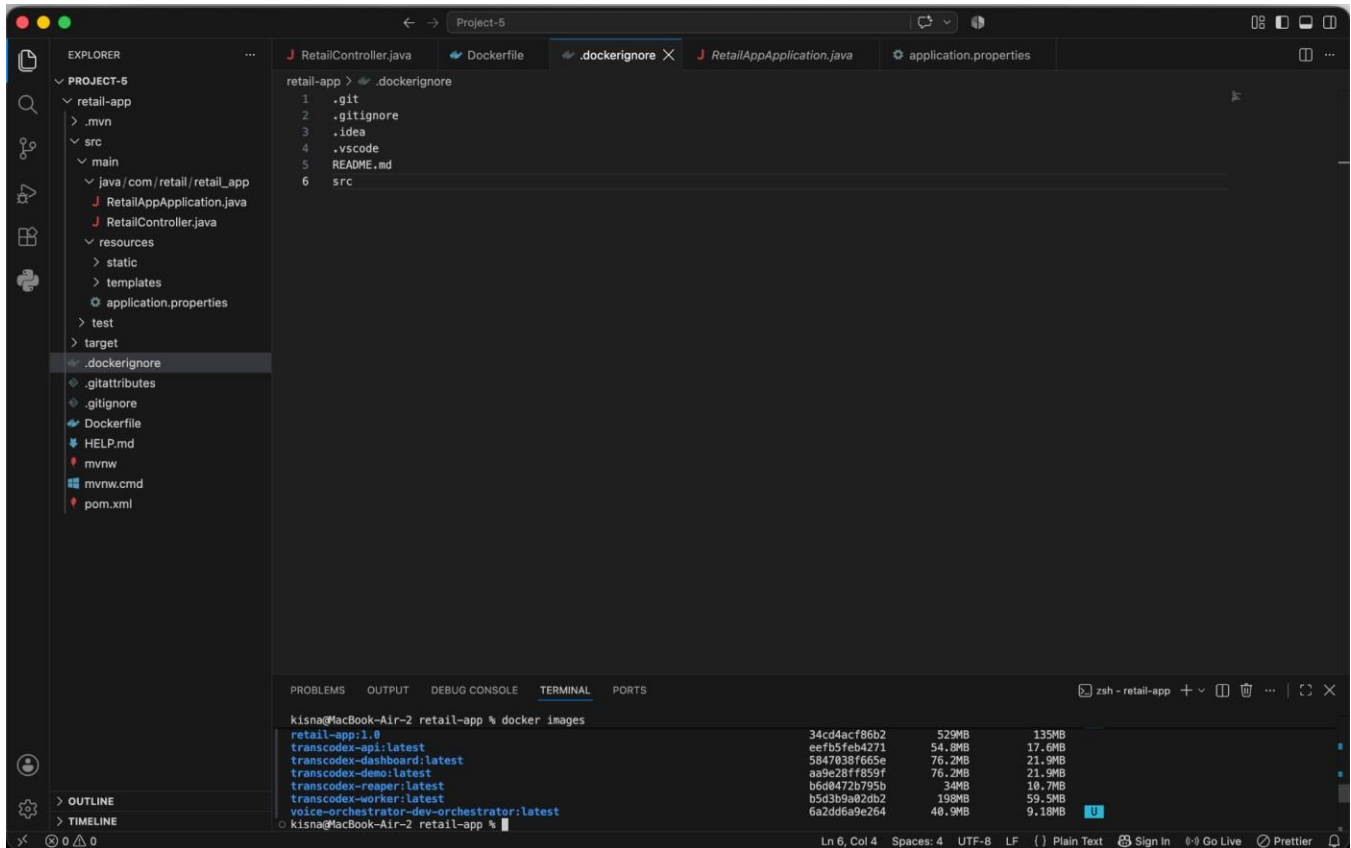
```
retail-app > .dockerignore
1 .git
2 .gitignore
3 .idea
4 .vscode
5 README.md
6 src

klsna@MacBook-Air-2 retail-app % clear
klsna@MacBook-Air-2 retail-app % docker build -t retail-app:1.0 .
[+] Building 13.4s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 184B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jre
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> transferring context: 83B
=> [1/3] FROM docker.io/library/eclipse-temurin:17-jre@sha256:e4f018a55645ad204892e44eb35437518d7e108ba2a2dce305024ab371d24876
=> resolve docker.io/library/eclipse-temurin:17-jre@sha256:e4f018a55645ad204892e44eb35437518d7e108ba2a2dce305024ab371d24876
=> sha256:44aa29c81f4b6e42893d918d7fe5c5066d20aed0e7733a331ae6754102d48030 2.28kB / 2.28kB
=> sha256:4449c111f6df84b0e5a1edc654c2ec9299973be7c8556d3b170b2f0d903c9eb8 159B / 159B
=> sha256:13f170f81eb47a9eb8a0f0b3533dc343bdff1714aba8f8d3da084ce7ce656a2 47.05MB / 47.05MB
=> sha256:f30e1adfc3e0182b8592b3c73667a78ec9434becd3f2ead4a2754866a3ca625f 29.39MB / 29.39MB
=> sha256:557836a62b7606e93caad19a768fa87fc346bf7d816ac03108c09adb1d684b 388B / 388B
=> sha256:d73407a274f66cb1e1017aa5257558853ec6b07340173b539ca22894 40.70MB / 40.70MB
=> extracting sha256:d73407a274f66cb1e1017aa5257558853ec6b07340173b539ca22894 0.45s
=> extracting sha256:557836a62b7606e93caad19a768fa87fc346bf7d816ac03108c09adb1d684b 0.05s
=> extracting sha256:f30e1adfc3e0182b8592b3c73667a78ec9434becd3f2ead4a2754866a3ca625f 0.25s
=> extracting sha256:13f170f81eb47a9eb8a0f0b3533dc343bdff1714aba8f8d3da084ce7ce656a2 0.45s
=> extracting sha256:4449c111f6df84b0e5a1edc654c2ec9299973be7c8556d3b170b2f0d903c9eb8 0.05s
=> extracting sha256:40aa29c81f4b6e42893d918d7fe5c5066d20aed0e7733a331ae6754102d48030 0.25s
=> [internal] load build context
=> transferring context: 19.87MB
=> [2/3] WORKDIR /app
=> [3/3] COPY target/retail-app-0.0.1-SNAPSHOT.jar app.jar
=> exporting to image
=> exporting layers
=> exporting manifest sha256:f120408bb3305dcccba934cddc6e5bf7b4f4a6224325beb650198c2278e0da99 0.05s
=> exporting config sha256:a8ec5bb75a60f8c7176309a214ab313e6180dc20e751d267ac68c0807e9371a4 0.05s
=> exporting attestation manifest sha256:ac1f67f1c36c234d08401346c662791c3d2539fe0a4570143a0117f4440bcf 0.05s
=> exporting manifest list sha256:34cd4acf86b244cc6ae84cd0d241d81a8f17cdf25370be1f8f53fba5f866698 0.05s
=> naming to docker.io/library/retail-app:1.0
=> unpacking to docker.io/library/retail-app:1.0
klsna@MacBook-Air-2 retail-app %
```

Figure 4: Docker Image Successfully Built

5. Docker Image Verified with docker images

This screenshot verifies that the retail-app:1.0 Docker image exists locally and displays its image ID and size.



The screenshot shows a Visual Studio Code editor window with a project named 'Project-5'. The Explorer sidebar on the left shows the project structure, including a 'retail-app' directory with subdirectories like 'main', 'resources', 'static', 'templates', 'test', and 'target'. The main editor area displays the '.dockerignore' file with the following content:

```
1 .git
2 .gitignore
3 .idea
4 .vscode
5 README.md
6 src
```

The bottom panel shows the 'TERMINAL' tab with the command 'docker images' executed in a shell. The output lists several Docker images, including 'retail-app:1.0'.

Image Name	Image ID	Size	Virtual Size
retail-app:1.0	34cd4ac86b2	529MB	135MB
transcodex-api:latest	eefb5feb4271	54.8MB	17.6MB
transcodex-dashboard:latest	5847038f665e	76.2MB	21.9MB
transcodex-demo:latest	aa9e28ff859f	76.2MB	21.9MB
transcodex-reaper:latest	b6d0472b795b	34MB	10.7MB
transcodex-worker:latest	b5d3b9a02db2	198MB	59.5MB
voice-orchestrator-dev-orchestrator:latest	6a2dd6a9e264	40.9MB	9.18MB

Figure 5: Docker Image Verified with docker images

6. Retail Application Container Running

This screenshot shows the retail-container running from the retail-app:1.0 image with port 8080 mapped from the host to the container.

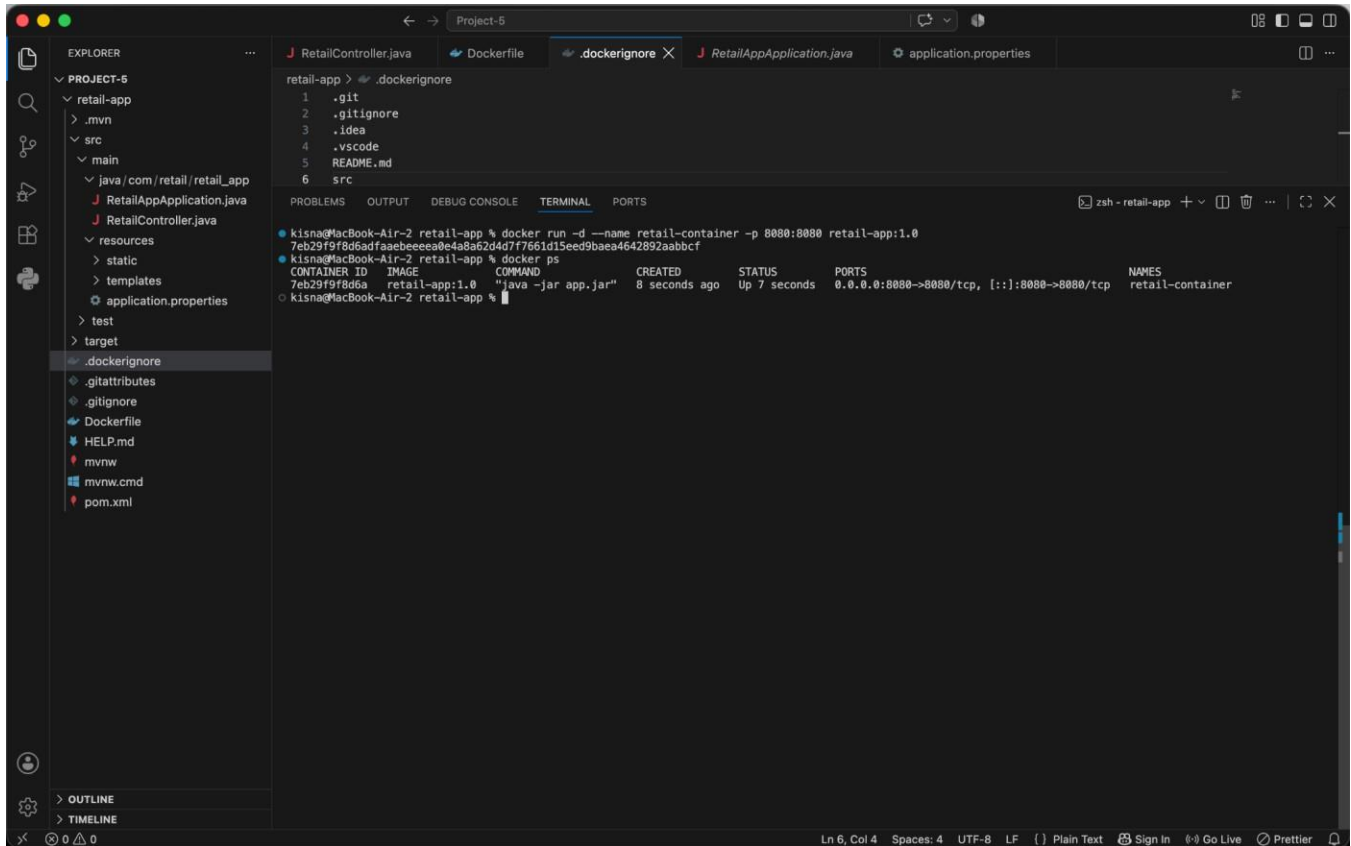


Figure 6: Retail Application Container Running

7. Dockerized Retail Application Verified in Browser

This screenshot confirms that the Spring Boot application is successfully running inside the Docker container and is accessible through localhost:8080.

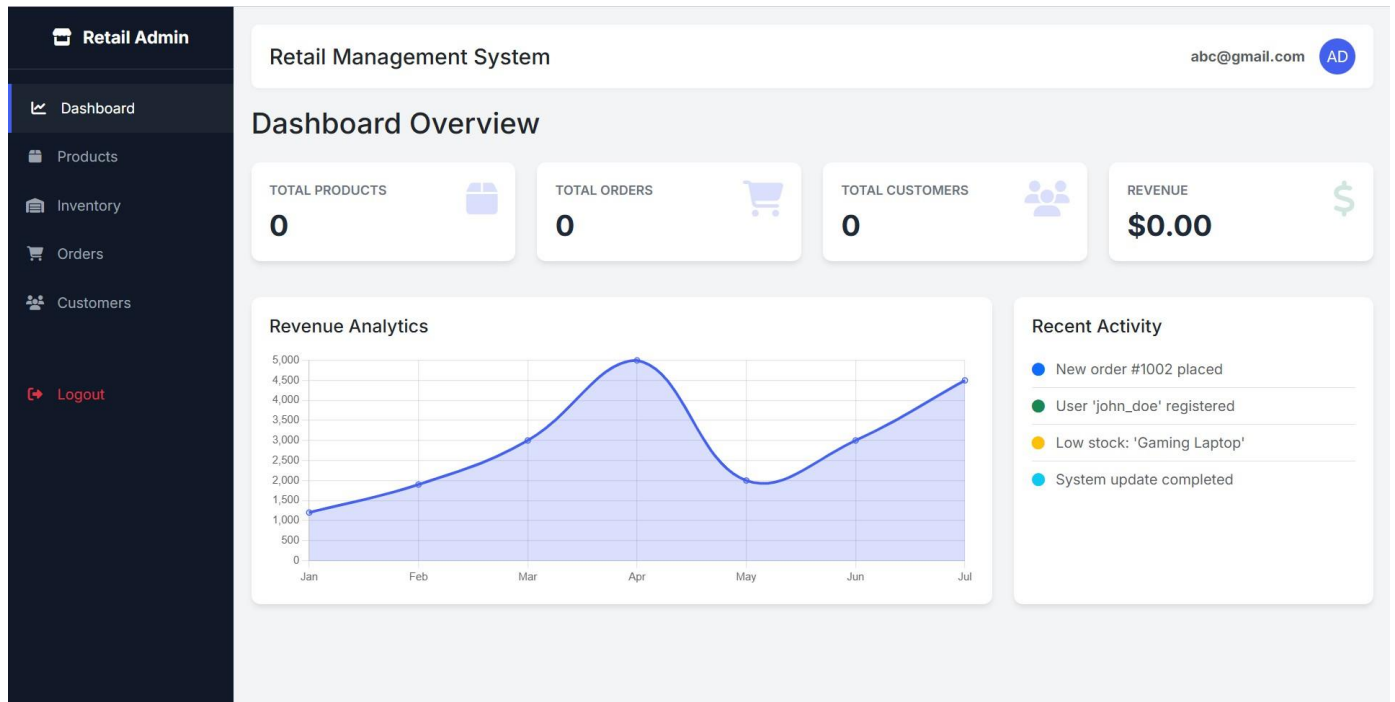
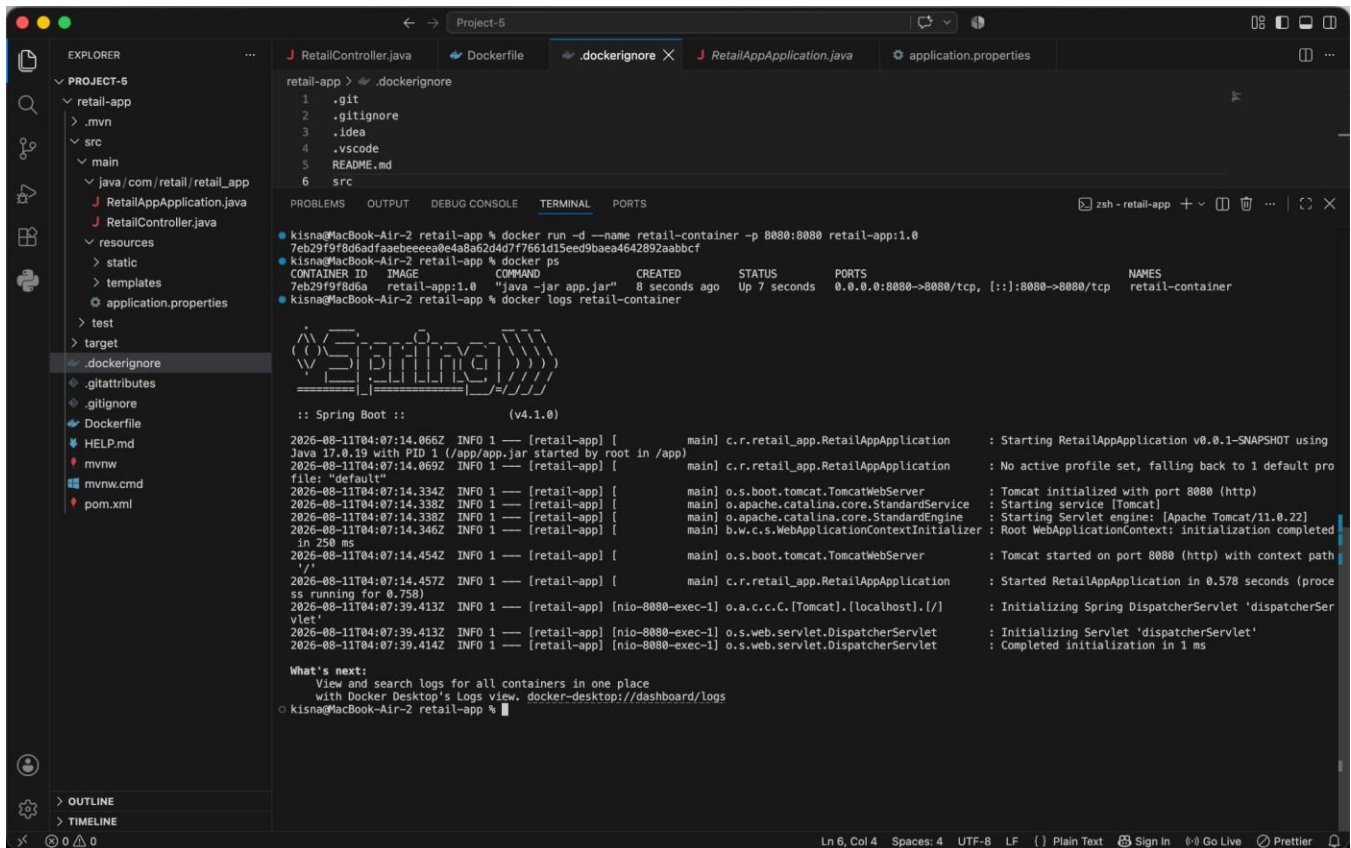


Figure 7: Dockerized Retail Application Verified in Browser

8. Docker Container Logs Showing Successful Startup

This screenshot displays the container logs, confirming that Spring Boot started successfully and Tomcat is listening on port 8080.



```
retail-app > .dockerignore
1 .git
2 .gitignore
3 .idea
4 .vscode
5 README.md
6 src

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
kiswa@MacBook-Air-2 retail-app % docker run -d --name retail-container -p 8080:8080 retail-app:1.0
7eb29f9f8d6adfaebee0e4a8a62d4d7f7661d15eed9baea4642892aabbcf
kiswa@MacBook-Air-2 retail-app % docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
7eb29f9f8d6a retail-app:1.0 "java -jar app.jar" 8 seconds ago Up 7 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp retail-container
kiswa@MacBook-Air-2 retail-app % docker logs retail-container

:: Spring Boot :: (v4.1.0)

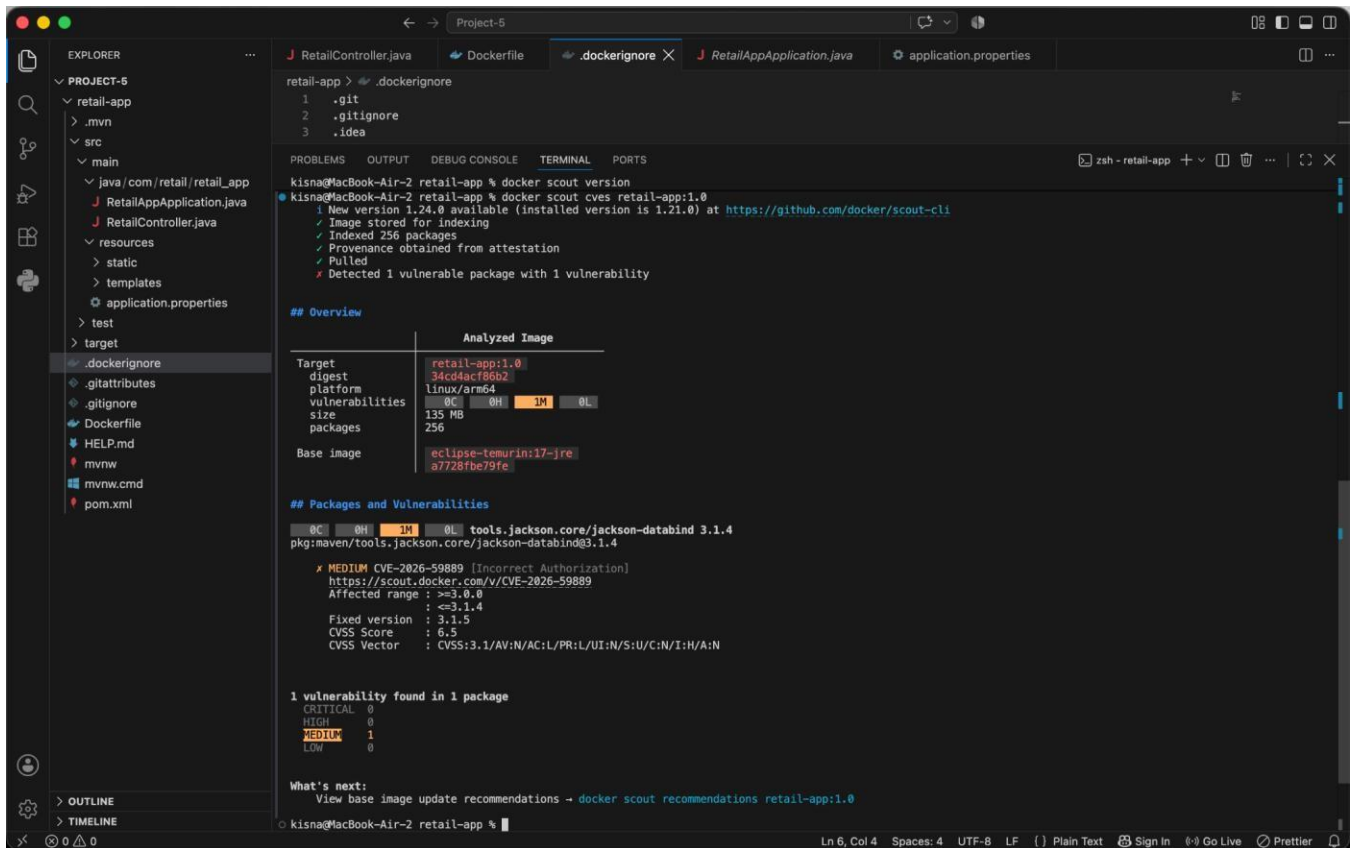
2026-08-11T04:07:14.066Z INFO 1 --- [retail-app] [ main] c.r.retail_app.RetailAppApplication : Starting RetailAppApplication v0.0.1-SNAPSHOT using
Java 17.0.19 with PID 1 (/app/app.jar started by root in /app)
2026-08-11T04:07:14.069Z INFO 1 --- [retail-app] [ main] c.r.retail_app.RetailAppApplication : No active profile set, falling back to 1 default pro
file: "default"
2026-08-11T04:07:14.334Z INFO 1 --- [retail-app] [ main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-08-11T04:07:14.338Z INFO 1 --- [retail-app] [ main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-08-11T04:07:14.338Z INFO 1 --- [retail-app] [ main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-08-11T04:07:14.346Z INFO 1 --- [retail-app] [ main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed
in 250 ms
2026-08-11T04:07:14.454Z INFO 1 --- [retail-app] [ main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path
"/"
2026-08-11T04:07:14.457Z INFO 1 --- [retail-app] [ main] c.r.retail_app.RetailAppApplication : Started RetailAppApplication in 0.578 seconds (proce
ss running for 0.758)
2026-08-11T04:07:39.413Z INFO 1 --- [retail-app] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherSer
vlet'
2026-08-11T04:07:39.413Z INFO 1 --- [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-08-11T04:07:39.414Z INFO 1 --- [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms

What's next:
View and search logs for all containers in one place
with Docker Desktop's Logs view. docker-desktop:///dashboard/logs
kiswa@MacBook-Air-2 retail-app %
```

Figure 8: Docker Container Logs Showing Successful Startup

9. Docker Scout Vulnerability Scan

This screenshot shows Docker Scout scanning the retail-app:1.0 image and identifying one medium-severity vulnerability in a Jackson databind package.



```
retail-app > .dockerignore
1 .git
2 .gitignore
3 .idea

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
kiswa@MacBook-Air-2 retail-app % docker scout version
kiswa@MacBook-Air-2 retail-app % docker scout cves retail-app:1.0
1 New version 1.24.0 available (installed version is 1.21.0) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 256 packages
✓ Provenance obtained from attestation
✓ Pulled
✗ Detected 1 vulnerable package with 1 vulnerability

## Overview

Analyzed Image
Target retail-app:1.0
digest 34cd4acf86b2
platform linux/arm64
vulnerabilities 0C 0H 1M 0L
size 135 MB
packages 256
Base image eclipse-temurin:17-jre
a7728fbc79fe

## Packages and Vulnerabilities
0C 0H 1M 0L tools.jackson.core/jackson-databind 3.1.4
pkg:maven/tools.jackson.core/jackson-databind@3.1.4

✗ MEDIUM CVE-2026-59889 [Incorrect Authorization]
https://scout.docker.com/v/CVE-2026-59889
Affected range : >=3.0.0
Fixed version : 3.1.5
CVSS Score : 6.5
CVSS Vector : CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:N/I:H/A:N

1 vulnerability found in 1 package
CRITICAL 0
HIGH 0
MEDIUM 1
LOW 0

What's next:
View base image update recommendations -> docker scout recommendations retail-app:1.0
kiswa@MacBook-Air-2 retail-app %
```

Figure 9: Docker Scout Vulnerability Scan

10. Docker Image Layer History

This screenshot displays the Docker image layer history, showing the Dockerfile instructions and layers used to construct the retail-app:1.0 image.

The screenshot shows an IDE window with the following components:

- EXPLORER:** A file tree on the left showing the project structure. The file `.dockerignore` is selected.
- Terminal:** A terminal window at the bottom showing the command `docker history retail-app:1.0` and its output.
- Table:** A table displaying the Docker image layer history. The table has columns for IMAGE, CREATED, CREATED BY, SIZE, and COMMENT.

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
34cd4acf86b2	6 minutes ago	ENTRYPOINT ["java" "-jar" "app.jar"]	0B	buildkit.dockerfile.v0
<missing>	6 minutes ago	EXPOSE [8080/tcp]	0B	buildkit.dockerfile.v0
<missing>	6 minutes ago	COPY target/retail-app-0.0.1-SNAPSHOT.jar ap...	19.9MB	buildkit.dockerfile.v0
<missing>	6 minutes ago	WORKDIR /app	8.19kB	buildkit.dockerfile.v0
<missing>	7 days ago	ENTRYPOINT ["/_cacert_entrypoint.sh"]	0B	buildkit.dockerfile.v0
<missing>	7 days ago	COPY --chmod=755 entrypoint.sh /_cacert_ent...	12.3kB	buildkit.dockerfile.v0
<missing>	7 days ago	RUN /bin/sh -c set -eux; echo "Verifying...	12.3kB	buildkit.dockerfile.v0
<missing>	7 days ago	RUN /bin/sh -c set -eux; ARCH="\$(dpkg --g...	143MB	buildkit.dockerfile.v0
<missing>	7 days ago	ENV JAVA_VERSION=jdk-17.0.19+10	0B	buildkit.dockerfile.v0
<missing>	7 days ago	RUN /bin/sh -c set -eux; apt-get update;...	95.6MB	buildkit.dockerfile.v0
<missing>	7 days ago	ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_A...	0B	buildkit.dockerfile.v0
<missing>	7 days ago	ENV PATH=/opt/java/openjdk/bin:/usr/local/sb...	0B	buildkit.dockerfile.v0
<missing>	7 days ago	ENV JAVA_HOME=/opt/java/openjdk	0B	buildkit.dockerfile.v0
<missing>	2 weeks ago	umoci raw add-layer --image /home/build/roc...	12.3kB	Add rock control metadata
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set annotations
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set labels
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set default PATH for bare-based rock
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set default commands
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set entrypoint
<missing>	2 weeks ago	umoci raw add-layer --image /home/build/roc...	135MB	

Figure 10: Docker Image Layer History